

Synthetic Loan-Application Data Generation and Machine-Learning Models for Credit-Risk Grading and Approval Decisions

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Synthetic Data and Modelling Pipeline
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Abstract—Real consumer-lending data is restricted by privacy law and commercial confidentiality, which makes it difficult to build and openly demonstrate credit models. This paper presents a fully reproducible pipeline that *generates realistic synthetic loan-application data* and trains machine-learning models on it to predict two outcomes: an applicant’s *credit-risk grade* and the lender’s *approval decision*. The work progresses through two iterations: Version 1, a broad United-States-style dataset with 33 features, and Version 2, a focused dataset grounded in the Cambodian micro-lending market with 15 features. Both attach transparent, rule-based labels so that the ground truth is auditable rather than a black box. On a held-out test set, the risk model attains approximately 91% accuracy and the approval model approximately 78%; we show that the gap is a *deliberate* consequence of injecting realistic stochasticity into the approval rule, not a modelling deficiency. The deliverable is a config-driven codebase that cleanly separates data generation from model training and exports serveable pipelines.

Index Terms—synthetic data, credit risk, loan approval, machine learning, random forest, reproducibility, micro-finance.

I. INTRODUCTION

Lenders must decide (a) how risky a borrower is and (b) whether to approve a requested loan. Building machine-learning models for these tasks normally requires historical loan books, which are private. Without data, one cannot prototype, teach, or interview around the problem. The central question of this work is therefore: *can we synthesise data that is realistic enough to support meaningful credit modelling, while keeping the labelling logic fully transparent?*

A. Objectives

- 1) Generate a synthetic loan-application dataset that encodes plausible socio-economic correlations (income, region, employment, loan products).
- 2) Attach rule-based labels for a *risk grade* and an *approval status*.
- 3) Train and evaluate models that recover those labels.
- 4) Make the pipeline reproducible and production-shaped, exporting saved pipelines that can be served from an application.

B. Contributions

- A documented, seed-reproducible synthetic data generator for the Cambodian lending context.

- A transparent points-based risk scorecard and a probabilistic approval rule that together define the learning targets.
- Two trained `scikit-learn` pipelines and an evaluation harness.
- A software design that decouples data generation from training, the two communicating only through a documented CSV schema.

II. BACKGROUND AND MOTIVATION

A. Why synthetic data

Synthetic data sidesteps privacy and licensing constraints and lets us *control* the data-generating process. The trade-off is that a model trained on it only learns the assumptions baked into the generator. We mitigate this by keeping those assumptions explicit and inspectable in source code rather than hidden.

B. Why the Cambodian context

Version 1 used generic United-States-style ranges. Version 2 was re-grounded in the Cambodian micro-finance market because this makes the synthetic ranges concrete and defensible: USD-denominated loans, province-level geography, product types common to Cambodian micro-finance institutions (moto, agriculture, small-and-medium-enterprise loans), and the high nominal interest rates typical of the sector (e.g. 24–36% on moto loans). The dataset then reflects a real domain rather than abstract noise.

III. THE TWO VERSIONS

The project evolved through two iterations, both preserved as reference notebooks. Table I summarises the differences.

A. Version 1: broad feature exploration

Version 1 generated 40,000 records with 33 features spanning demographics (age, education, marital status, dependents), credit behaviour (credit score, utilisation, inquiries, payment history, bankruptcies), and balance-sheet items (assets, liabilities, net worth, savings). It derived an interest rate from credit score, amount and term, computed a total debt-to-income ratio, then applied a deterministic weighted-sum approval rule and a second weighted aggregation bucketed

TABLE I
COMPARISON OF THE TWO PROJECT VERSIONS.

Aspect	Version 1 (legacy)	Version 2 (productionised)
Market	US-style, generic	Cambodian micro-lending
Features	33	15
Targets	Approval (class.), Risk score (regression)	Risk category (3-class), Loan status (binary)
Risk label	Weighted sub-scores	Points scorecard → 3 bands
Approval	Deterministic threshold	Probabilistic, by risk band
Models	Logistic Reg., RF, Gradient Boosting	Random Forest pipelines
Saved?	No (notebook only)	Yes (joblib + metrics)

into Low/Medium/High risk. It trained Logistic Regression, Random Forest and Gradient Boosting models, reaching about 91% approval accuracy and $R^2 \approx 0.76$ on the risk-score regression. Its weakness was complexity: 33 features, several weakly motivated, and no model persistence.

B. Version 2: focused, grounded, productionised

Version 2 reduced the feature set to the 15 most decision-relevant variables and re-grounded them in Cambodia. It introduced the points-based scorecard for risk and a probabilistic approval rule, trained one Random Forest pipeline per target, and saved the pipelines for serving. This is the version rebuilt into the production package; Version 1 remains a reference.

IV. METHODOLOGY

A. Data-generating process

Each application is constructed in a fixed *causal order*, so downstream variables depend on upstream ones, producing realistic correlations rather than independent noise:

```
Province -> RegionType -> EmploymentType -> Income
                        | Age ____/
                        v
LoanType -> LoanAmount -> Term, Rate -> DTI
                        v
RiskScore -> RiskCategory -> LoanStatus
```

The principal conditional rules are: region drives the employment mix (rural areas skew toward farming, urban toward salaried work); income depends on region, employment and age (prime age 25–45 receives a higher ceiling); the loan product depends on the applicant segment; the loan amount is capped at five times annual income; and term and interest rate are product-specific. The debt-to-income ratio (DTI) uses the standard amortising-annuity payment

$$A = P \cdot \frac{r(1+r)^n}{(1+r)^n - 1}, \quad (1)$$

where P is the principal, r the periodic interest rate and n the number of periods; $DTI = A/\text{income}$.

TABLE II
RISK SCORECARD FACTOR WEIGHTS.

Factor	Max points
Employment type	25
Loan-to-income ratio	20
Loan purpose	15
Debt-to-income (DTI)	15
Credit history	12
Region	10

B. Labelling

Risk is a transparent scorecard in $[0, 100]$. Each factor contributes risk points and mitigating conditions subtract them; the maxima are given in Table II. The score is bucketed into Low (≤ 35), Medium (36–65) and High (> 65).

Approval is intentionally *probabilistic*: a base probability set by risk band (Low 85%, Medium 45%, High 15%) is adjusted by collateral, credit history, employment and product, after which a Bernoulli draw decides the outcome. This injects irreducible noise so the task is non-trivial.

C. Modelling

Both targets are learned by independent Random Forest classifiers wrapped in a `scikit-learn` pipeline:

```
ColumnTransformer(
    StandardScaler(numeric) +
    OneHotEncoder(categorical))
-> RandomForestClassifier(
    n_estimators=200, max_depth=12,
    class_weight="balanced")
```

Bundling preprocessing inside the pipeline guarantees train/serve consistency: the saved artifact consumes raw application records directly, with no separate feature-engineering step that could drift.

D. Experimental setup and leakage control

We generate 40,000 applications and use an 80/20 stratified train/test split with a fixed seed. We report accuracy, macro and weighted F_1 , the confusion matrix, and a per-class report. Crucially, the raw `RiskScore` (the intermediate that defines the risk band) and the `RiskCategory` label are *excluded* from the feature set; both models consume the same 15 raw features, so the saved pipelines never observe leakage columns.

V. RESULTS

On the held-out test set of 8,000 applications, the models achieve the scores in Table III. A representative risk confusion matrix (rows are true classes, columns predicted, order Low/Medium/High) is

$$\begin{bmatrix} 4061 & 278 & 0 \\ 171 & 2624 & 192 \\ 0 & 56 & 618 \end{bmatrix}.$$

Interpretation. The risk model is highly accurate because its label is a deterministic function of the features; a tree

TABLE III
HELD-OUT TEST-SET PERFORMANCE.

Model	Accuracy	Macro F_1
Risk (Low/Medium/High)	≈ 0.91	≈ 0.89
Approval (Approved/Reject)	≈ 0.78	≈ 0.76

ensemble recovers a scorecard well, and its errors concentrate on *adjacent* bands rather than skipping a band. The approval model plateaus near 0.78 *by design*: the approval label contains a Bernoulli coin-flip, so part of the outcome is irreducible. A model scoring near 0.99 here would instead signal label leakage or a broken deterministic rule.

VI. DISCUSSION

The project demonstrates a complete, reproducible workflow from data synthesis to a serveable model, with a disciplined separation between the generation and training packages, which share only a documented CSV contract. Evaluation is honest: the approval ceiling is explained rather than hidden. Engineering choices that support this include a single configuration file (sample size, distributions, features and hyper-parameters are data, not code), single-seed reproducibility, and an auditable ground-truth scorecard.

A. Limitations

The data is synthetic and encodes hand-tuned assumptions; it is not validated against a real loan book, and the distributions are designed rather than fitted to survey data. Sensitive-adjacent attributes (gender, province) are present for realism but no fairness audit has been performed; consequently the models must not be used for real lending decisions.

B. Future work

Promising extensions include calibrating distributions against published Cambodian micro-finance statistics, adding probability-calibrated outputs and tunable decision thresholds for the approval model, a fairness analysis across gender and region, gradient-boosting and logistic baselines for comparison, and unit tests for the scorecard and samplers.

VII. CONCLUSION

A carefully designed synthetic dataset can support a credible, end-to-end credit-modelling workflow without any real customer data. By grounding Version 2 in the Cambodian micro-lending market and keeping the labelling logic transparent, the pipeline produces models whose strengths (a highly learnable risk grade) and limits (an intentionally noisy approval task) are both explainable. The result is a reproducible, production-shaped codebase suitable for demonstration, teaching, and integration into a web application.

REPRODUCIBILITY

The full pipeline is reproduced with:

```

pip install -r requirements.txt
python -m data_generation.run
python -m model_training.train
python -m model_training.predict
# or simply: make all

```

REFERENCES

- [1] M. Mitchell *et al.*, “Model cards for model reporting,” in *Proc. Conf. Fairness, Accountability, and Transparency*, 2019, pp. 220–229.
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